

2026.03.17

TRPO: $\left\{ \begin{array}{l} \text{REINFORCE: On-policy} \Rightarrow \text{Off-policy} \\ \text{New policy after each update is close} \\ \text{to the original Policy} \end{array} \right.$

Policy Gradient: $\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta} \cdot A(a, s)]$

$$\begin{array}{c} \nearrow \theta + \alpha \cdot \nabla_{\theta} J(\theta) \\ \circ \\ \theta \end{array}$$

Trust Region Constraint

TRPO: $\max_{\theta} J_{\pi_{\theta+d}} \quad \text{s.t.} \quad \text{KL}(\pi_{\theta+d} \parallel \pi_{\theta}) \leq d$

$$g \triangleq \nabla_{\theta} J_{\pi_{\theta}} \quad H = \mathbb{E} [\nabla_{\theta} \log \pi_{\theta} \cdot \nabla_{\theta} \log \pi_{\theta}^T]$$

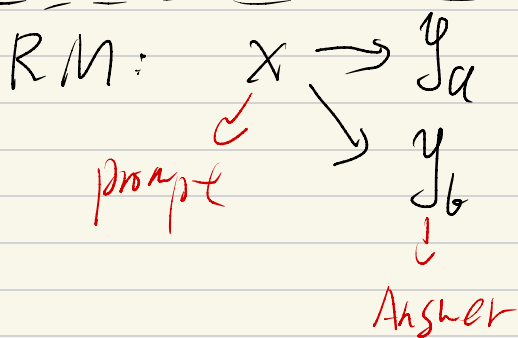
$$\max_x g^T \cdot x$$

$$\text{s.t.} \quad \frac{1}{2} x^T H \cdot x \leq d$$

PP0-Clip

$$\min \{ r(\theta) \cdot A, \text{Clip}\{r(\theta), 1-\epsilon, 1+\epsilon\} \cdot A \}$$

$$\text{Clip}\{r(\theta), 1-\epsilon, 1+\epsilon\} = \frac{r(\theta)}{1-\epsilon} \left[\frac{r(\theta)}{1+\epsilon} \right]$$



$r(y, x)$
reward model

$$\text{BTL: } P[y_a \succ y_b | x] = \frac{\exp(r(y_b, x))}{\exp(r(y_b, x)) + \exp(r(y_a, x))}$$

MLE:

$$\sum_w y_i \log(p) + \sum_{w_{i+1}} y_i \log(1-p)$$

$$= \text{Sigmoid}(r(y_a, x) - r(y_b, x))$$

$$\text{Sigmoid}(x) = \frac{1}{1 + \exp(-x)}$$